

Group 1

L02 ML development tools

ITAI-1371

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Module 2 Lab Reflection Journal

Before this lab, we had heard of pandas, NumPy, and Matplotlib, but we were not really sure how they worked together in a machine learning project. Using the Iris dataset helped us see how each tool connects and made the whole process easier to understand. The dataset was small and easy to work with, which helped us focus on learning the process without feeling overwhelmed by too much information. By the end of the lab, we understood the basic order much better: load the data, organize it, explore what it contains, calculate useful values, and create graphs to find patterns.

Pandas made the biggest difference in how we looked at the data. When we first saw the Iris measurements as an array, they looked like rows of random numbers. Turning them into a DataFrame gave every column a name and made the information feel more organized. Using `head()` let us check the first few flowers, while `info()` showed the data types and confirmed that all 150 rows had complete values. That step seemed simple, but it taught us an important habit. Before we start analyzing data, we should first check that the dataset has the right columns, values, and overall structure. Doing this early can help us avoid mistakes and problems later.

The NumPy part also helped us understand why summary statistics matter. The mean sepal length came out to about 5.84 centimeters, and the standard deviation came out to about 0.83 centimeters. Before this lab, we mostly saw the average as just the final result. Now we understand that the mean shows the center of the data, while the standard deviation shows how

much the measurements spread out around that center. Looking at both values gives a better picture than looking at either one alone. We also liked that NumPy could perform the calculation on the whole column without needing a loop. This made vectorized operations easier to understand and showed us how Python libraries can make data analysis much faster and simpler.

The scatter plot gave us the clearest learning moment. A table showed the exact measurements, but it was hard to notice patterns by looking at the numbers one row at a time. Once we plotted sepal length against sepal width and used a different color for each species, the pattern became much easier to see. Setosa stood out from the other species because it usually had shorter sepal lengths and wider sepals. Versicolor and Virginica overlapped much more, so those two measurements would not identify them as easily. That overlap helped us understand why feature choice matters. A model only learns from the information we give it, and some features separate groups better than others.

The graph also made us curious about the petal measurements. Since sepal length and width do not clearly separate Versicolor from Virginica, we would like to compare petal length and petal width next. The species averages make it seem like petal measurements could show a clearer difference. We would also like to try different feature combinations and see which ones predict the species most accurately. This was one of the first times we saw a visualization create new questions instead of only presenting an answer.

The species count bar chart helped us notice something we had not really thought about before, which is class balance. The dataset contains 50 Setosa, 50 Versicolor, and 50 Virginica samples. Because each class has the same number of examples, one species does not dominate the dataset. We can now see how an unbalanced dataset could mislead a model. If most of the flowers came from only one species, a model could keep guessing that species and still look

accurate. Checking the class counts before training a model gives us a better idea of how to understand its results.

Overall, this lab made the beginning of a machine learning workflow feel less intimidating. Pandas helped us organize and inspect the data, NumPy helped us summarize it, and Matplotlib helped us see relationships that the raw numbers hid. Moreover, we learned that exploring the data is not something we should rush through before building a model. It can help us find problems, choose better features, and think of new questions to explore. Working in a notebook made the process easier because we could run one section at a time, check the output, and fix mistakes before moving on. We still need more practice with choosing the right graphs and understanding statistics, but now we have a much better idea of how these tools work together in machine learning.